Michele Valsecchi

Materials Scientist · Polymer Physics · Soft Matter

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PROFESSIONAL SUMMARY

Materials scientist/engineer specialized in complex fluids and soft matter. I combine theory, computational physics and experiments to develop physical insight and help design novel advanced materials. I am passionate about collaborative, interdisciplinary and international science.

RESEARCH EXPERIENCE

Columbia University · Postdoctoral Research Scientist — 2023 — Present

Collaborate with polymer scientists and engineers across institutions in the US, Europe and India. Co-supervised, trained and mentored 8 PhD and 10+ BSc/MSc students; helped write 2 grant proposals (1 internal, 1 NSF) and led 5 research projects.

Funding

- Department of Energy — Genesis Mission: “AI-Enabled Physical Operating System for Bio-programmable Matter” (2026–2027), PI Prof. Kyle Bishop.
- National Science Foundation #2225290: “Multiscale Data-Physics Models for the Critical Role of Interfaces in Overmolded Thermoplastic Parts” (2023–2026), PI Prof. Sanat Kumar.

Active research areas

- Multiphysics simulation of plastics processing, DNA self-assembly and gel thermodynamics.
- Polymer degradation, with a focus on nanoplastics release.
- Statistical mechanics of polymers and complex fluids in and out of equilibrium.
- Closed-loop AI-driven experimentation.

TECHNICAL SKILLS

Theory Statistical mechanics Polymer physics Condensed matter Self-consistent field theory	Computational physics Molecular dynamics (LAMMPS) Monte Carlo Density functional theory (PySCF)	Programming Python · HPC/SLURM Unix/Bash · Git · Julia	Experimental SAXS/WAXS · DSC/TGA GC–MS · FTIR · Raman

EDUCATION

Imperial College London · PhD in Theory and Simulation of Materials — 2019 — 2023

Fully funded by Procter & Gamble. Member of the Thomas Young Centre and the Centre for Process Systems Engineering. Supervisors: Prof. George Jackson and Prof. Amparo Galindo. Thesis: “Molecular modelling and statistical thermodynamics of semi-crystalline polymers: network constraints, fluid sorption and surface melting”. Applications to packaging, drug excipient and reactor mixture design. Wrote 1 EPSRC grant to optimize polymer liners for H_2 storage.

Imperial College London · MSc in Theory and Simulation of Materials — 2018 — 2019

Distinction. Thesis: “Modelling the thermodynamic properties of aqueous PEG mixtures”. Focus on computational physics (MD, MC, Free Energy Methods, DFT, AI/ML) and grant writing.

Stanford University · Undergraduate Summer Visitor, Organic Chemistry Intensive — Summer 2017

Grade A+ (top 1%). Experience with polymer synthesis and analytical techniques.

Politecnico di Milano · BSc in Materials and Nanotechnology Engineering — 2015 — 2018

110/110 cum laude; valedictorian. Thesis: “Anodic intercalation of graphite: application to the production of graphene and graphene oxide-based systems”.

TEACHING

Columbia University · Lecturer, Polymer Physics — 2024 — 2025

Co-designed and delivered the graduate Polymer Physics course in the Department of Chemical Engineering. Prepared original lecture material and examined about 20 students.

Imperial College London · Graduate Teaching Assistant — 2018 — 2023

400+ h marking, tutoring and mentoring for courses in the Materials and Chemical Engineering Departments. Outstanding GTA Student Choice Award (2021).

Associate Fellow of Advance HE since 2023.

PUBLICATIONS

1. Valsecchi M, Ninarello A, Zaccarelli E, Kumar SK. “Phase transitions of polymer networks under tension.” In press in Macromolecules (2026).
2. Pai V, Kalghatgi SR, Krol M, Valsecchi M, Mendez NF, Banks C, Ruokolainen J, Müller AJ, Kumar SK, Schadler LS. “Acoustic field–assisted nanoparticle ordering in poly(ethylene oxide) nanocomposites.” In press in ACS Macro Letters (2026).
3. Valsecchi M, Fall WS, Meyer H, Grest GS, Kumar SK. “Molecular insights on the thermal welding of semicrystalline polymers.” Physical Review Letters (2026), 137, 098101.
4. Valsecchi M, Kalghatgi SR, Mendez NF, Lincoln C, Rorrer NA, Kumar SK. “Role of amorphous chains in nanoplastics formation from semicrystalline polymers.” ACS Macro Letters (2026), 15(9), 1228–1232.
5. Liu T, Cui J, Kantert A, Tabib M, Kopic N, Valsecchi M, Kumar SK, Fish J, Simon J-W. “Thermomechanically coupled visco–hyperelastic–plastic model with self-heating for thermoplastics.” Computer Methods in Applied Mechanics and Engineering (2026), 461, 119127.
6. Kots PA, Hinton ZR, Zare M, Vance BC, Ley-Flores M, de Pablo JJ, Epps TH, Korley LTJ, Valsecchi M, Jackson G, Galindo A, Vlachos DG. “Impact of small-alkane solvents on polyolefin hydrogenolysis over a ruthenium catalyst.” Industrial & Engineering Chemistry Research (2026), 65(18), 9410.
7. Singh M, Mendez NF, Valsecchi M, Kumaraswamy G, Kumar SK. “Materials science underpinnings of micro and nanoplastics.” Soft Matter (2025), 21(30), 6023–6033.
8. Mendez NF, Sharma V, Valsecchi M, Pai V, Lee JK, Schadler LS, Müller AJ, Watson-Sanders S, Dadmun M, Kumaraswamy G, Kumar SK. “Mechanisms of quiescent nanoplastics formation in semicrystalline polymers.” Nature Communications (2025), 16(1), 3051.
9. Guo G, Saidi TL, Terban MW, Valsecchi M, Billinge SJL, Lipson H. “Ab initio structure solutions from nanocrystalline powder diffraction data via diffusion models.” Nature Materials (2025), 24, 1502.
10. Valsecchi M, Kumar SK. “Impact of chain ends on the amorphous phase topology in semicrystalline polymers.” Macromolecules (2025), 58(18), 10143–10163.
11. Cui J, Liu T, Valsecchi M, Giersberg M, Çelik H, Simon J-W, Kumar SK, Petersen J, Fish J. “Thermodynamically consistent coupled chemo-thermo-mechanical model of interfaces in overmolded thermoplastic parts.” Computer Methods in Applied Mechanics and Engineering (2025), 447, 118359.
12. Karmakar R, Venkatareddy N, Himanshu, Valsecchi M, Maiti PK, Sastry S, Kumar SK, Patra TK. “Computer simulations of entropic cohesion in reversibly crosslinked polymers.” Soft Matter (2025), 21(3), 348–355.
13. Chiapasco M, Valsecchi M, Hill G, Wallis C, Porter AE, Giuliani F. “Impacts of photooxidation on commercially available homo and copolymers of polypropylene on their microstructure and mechanics: a cross-sectional analysis.” Nano Select (2025), 6(1), e202300082.
14. Nguyen L, Valsecchi M, et al. “Modelling the solubility of CO_2 in rubbery and glassy amorphous PS and PMMA with the SAFT-γ Mie group-contribution EoS and the NET-GP approach.” Fluid Phase Equilibria (2025), 597, 114445.
15. Valsecchi M, Galindo A, Jackson G. “Modelling the thermodynamic properties of the mixture of water and polyethylene glycol (PEG) with the SAFT-γ Mie group-contribution approach.” Fluid Phase Equilibria (2024), 577, 113952.
16. Valsecchi M, Galindo A, Jackson G. “Modelling sorption of hydrocarbons in polyethylene with the SAFT-γ Mie approach combined with a statistical-mechanical model to describe semicrystalline polymers.” Macromolecules (2023), 56(24), 9952–9968.
17. Valsecchi M, Ramadani J, Williams D, Galindo A, Jackson G. “Influence of tie-molecules and microstructure on the fluid solubility in semicrystalline polymers.” The Journal of Physical Chemistry B (2022), 126(44), 9059–9088.

INVITED AND CONTRIBUTED TALKS

- Talk & poster, ESAT 2026 (Lisbon, Portugal — May 2026).
- Talk & poster, APS Global Physics Summit 2026 (Denver, CO — March 2026).
- Talk, APS March Meeting (Anaheim, CA — March 2025).
- Invited talk, Institute Charles Sadron (Strasbourg, France — December 2024).
- Talk, AIChE 2022 (Phoenix, AZ — November 2022).
- Talk, Thermodynamics 2022 (Bath, UK — September 2022).
- Talk & poster, ESAT 2022 (Graz, Austria — July 2022).

PRIZES AND AWARDS

- Julia Higgins Centenary Prize — Best PhD thesis in Chemical Engineering, Imperial College London (2023).
- Johnson Matthey Prize — Best PhD thesis in Theory and Simulation of Materials, Imperial College London (2023).
- Outstanding GTA Student Choice Award, Imperial College Union (2021) — one graduate teaching assistant selected each year by students.
- Best Freshmen of the A.Y. 2015/2016, Politecnico di Milano (2017).

MEMBERSHIPS AND QUALIFICATIONS

Advance HE — Associate Fellow; Royal Society of Chemistry — Associate Member; Institute of Physics — Member.

SERVICE

- Reviewer for 8 papers submitted to journals including Molecular Physics, Physical Review E and Macromolecules.
- Active member of the Union for Postdoctoral Researchers at Columbia CPW-UAW 4100.
- Volunteer student demonstrator at the Great Exhibition Road Festival, Imperial College London (2019).

References available upon request.